

Table 1: Comparison of clustering inertia for KSKM variants with and without centroid perturbation, reported as multiples of the baseline KSKM across datasets for successful runs.

Dataset	Method Constraint Level	KSKM	KSKM_repo	KSKM_E	KSKM_E_repo
COL1	0.05	1.000	1.001	1.000	1.002
	0.1	1.000	1.003	0.998	1.001
	0.15	1.000	1.005	0.999	1.001
	0.2	1.000	1.002	0.998	1.000
COL2	0.05	1.000	1.019	0.999	1.017
	0.1	1.000	1.023	1.001	1.018
	0.15	1.000	1.026	0.998	1.018
	0.2	1.000	1.027	1.003	1.014
	0.25	1.000	1.030	0.990	1.006
	0.3	1.000	1.035	0.977	0.991
	0.35	1.000	1.018	1.005	1.009
	0.4	1.000	1.021	1.001	1.013
	0.45	1.000	1.011	1.002	1.006
	0.5	1.000	1.009	1.002	1.002